Documentation

of

Project Report of Offa LGA Land Surface Temperature

using

2014 & 2024 Landsat 8 Data

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CHAPTER ONE

INTRODUCTION

1.1 Background Study

Rapid urban growth is a defining characteristic of settlements in many developing countries. This expansion is often reflected in population increases, land development, unregulated construction, and the depletion of vegetation due to the spread of impervious surfaces. As urban areas continue to expand, they undergo significant microclimatic changes, leading to the formation of **Urban Heat Islands (UHI)**—a phenomenon where urban centers experience elevated temperatures compared to their rural surroundings.

The **increase in atmospheric and surface temperatures** within urban environments is primarily driven by the replacement of natural green spaces with non-evaporative, heat-absorbing materials such as concrete, asphalt, and metal structures. These materials possess **high heat capacity and low solar reflectivity**, causing them to absorb and retain more heat, thereby intensifying surface temperatures. Areas with **poor greening** experience the most severe temperature rises, exacerbating the effects of urban heating.

This rise in urban temperatures has serious environmental and socio-economic consequences. Higher temperatures contribute to **increased energy demands** for cooling, elevate **air pollution levels**, and worsen **thermal discomfort** among urban dwellers. Additionally, prolonged exposure to high temperatures can lead to adverse health effects, including **heat strokes, dehydration, and physiological stress**, particularly among vulnerable populations.

Land Surface Temperature (LST) analysis is crucial for understanding the thermal variations across urban landscapes. By leveraging **remote sensing and GIS technologies**, researchers can monitor temperature trends, assess the impact of urban expansion on heat distribution, and develop mitigation strategies such as **urban greening, reflective roofing, and improved land-use planning**. Understanding LST dynamics is essential for promoting sustainable urban development and enhancing climate resilience in rapidly growing cities. Role of Remote Sensing in LST Analysis Using Satellite Imagery

Roles of Remote Sensing in Land Surface Temperature using Satellites imageries

Remote sensing provides an efficient and cost-effective method for estimating LST across large geographical areas using satellite imagery. Some key aspects include:

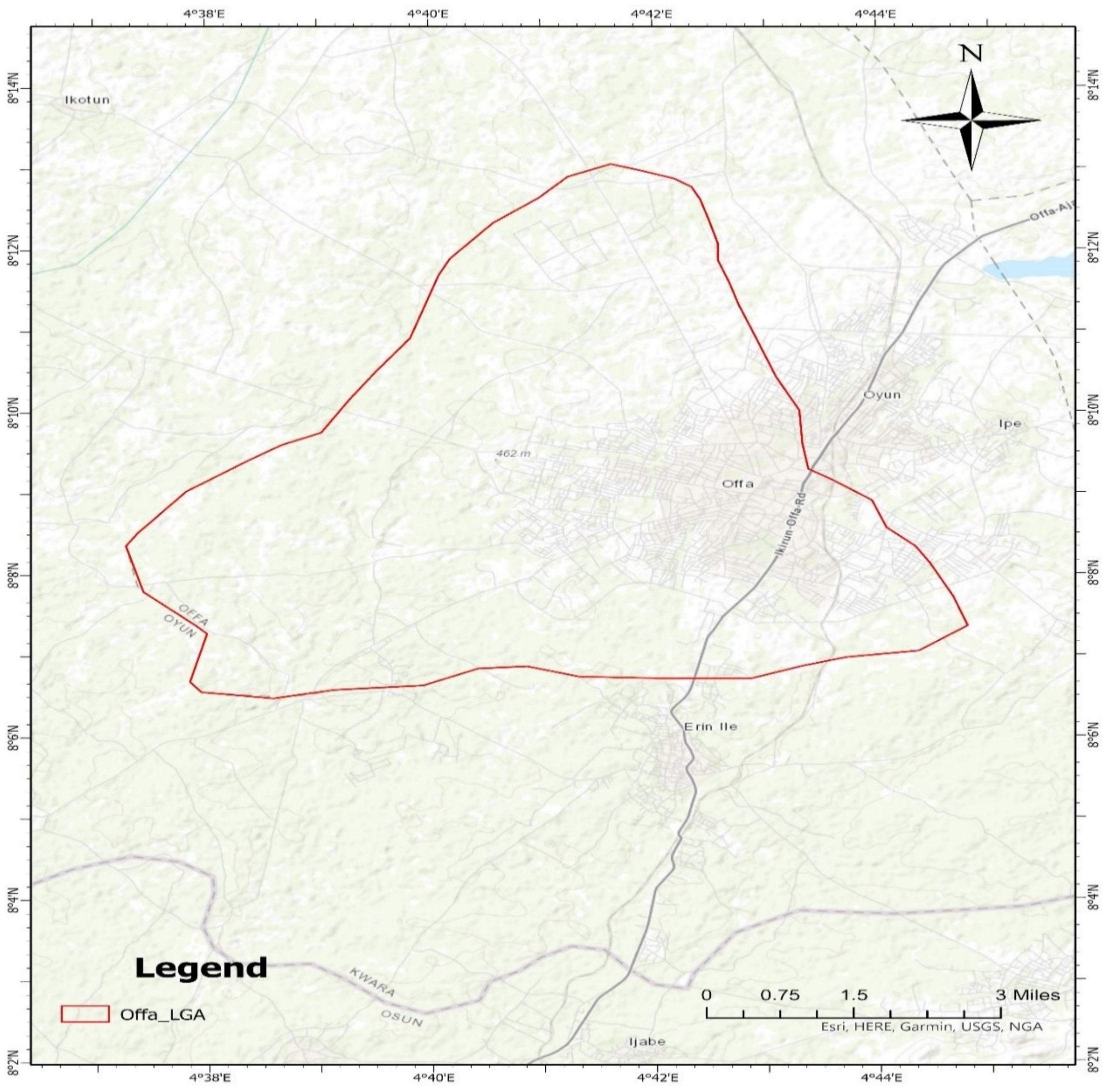
- **Satellite-Based Data Acquisition**: Satellites such as **Landsat 8/9**, MODIS, and Sentinel-3 provide thermal infrared (TIR) bands that capture surface temperature variations.
- **Multi-Temporal Analysis**: Remote sensing allows for continuous monitoring of LST changes over time, making it useful for studying seasonal and long-term climatic patterns.
- **Accurate and Scalable Measurements**: Unlike traditional ground-based temperature measurements, which are limited to specific locations, satellite-based remote sensing offers **spatially comprehensive and globally consistent** LST datasets.
- **Integration with GIS**: Geographic Information Systems (GIS) enhance LST analysis by enabling spatial visualization, classification, and statistical correlation with other environmental factors like vegetation cover, land use, and topography.

1.2 STUDY AREA

Offa local government was created in 1991 with headquarters at Offa. The local government area has 10 wards. It has a population of 90,000 inhabitants at the 2006 census. The major towns are Adeleke, Igbewere, Igbadu, and Offa. Yoruba is the predominant language spoken at the Offa LGA.

Offa is well known for cultivation of Sweet potatoes and maize which also formed part of the favourite staple foods of the indigenes in the town. Offa in one of her eulogy is being addressed as the home of sweet potatoes. It also has large deposits of Granite. Cattle, goats and sheep are also raised in the environs. The key religions practiced in the town are:- Islam, Christianity and Traditional religions.

Its major festivals are Egungun, and Moremi festival. Tourist attractions include the Onimokas Shrine Offa and the Olofa of Offa's palace.



STUDY AREA MAP
OFFA LGA

1.3 Aims and Objectives of the study

The aim of this study is to analyze **Land Surface Temperature (LST)** variations in the Offa LGA using **remote sensing and GIS techniques** to assess the impact of urbanization and environmental factors on surface temperature distribution. The study seeks to provide insights into the relationship between land cover changes and thermal characteristics, aiding in climate resilience and sustainable urban planning.

Objectives

To achieve this aim, the study is guided by the following objectives:

- **To preprocess and analyze Landsat satellite imagery** for accurate extraction of thermal infrared data relevant to LST computation.
- **To derive and compare LST values** using different spectral bands (Band 10 and Band 11) and validate the results through statistical analysis.
- **To assess the spatial distribution of LST** across different land cover types, identifying temperature variations associated with vegetation, built-up areas, and water bodies.
- **To investigate the impact of urbanization on LST patterns**, highlighting areas affected by the Urban Heat Island (UHI) effect.
- **To integrate LST analysis with GIS** for effective visualization and spatial interpretation of temperature trends.
- **To provide recommendations for urban planning and climate adaptation strategies** based on the findings of the LST analysis.

CHAPTER TWO

DATA AND METHODOLOGY

2.1 Data Collection

The data for this study was obtained from the United States Geological Survey (USGS) Earth Explorer platform, where Landsat Level 1 satellite imagery was downloaded for analysis. Landsat data was selected due to its high-resolution thermal infrared bands, which are essential for **Land Surface Temperature (LST) analysis**. The downloaded dataset included Thermal Infrared Sensor (TIRS) bands (Band 10 and Band 11) and (Band 4 and Band 5 for NDVI), which are commonly used for LST estimation.

The data used for the Land Surface Temperature data is multi-date satellites imageries (2014, 2024) covering the Offa local government area.

To ensure the accuracy and relevance of the study, careful attention was given to selecting a cloud-free or minimal cloud cover scene that fully covers **Offa Local Government Area (LGA)**. This step was crucial in obtaining clear and precise temperature readings without interference from atmospheric distortions. Additionally, metadata files accompanying the Landsat images were used to extract necessary radiometric and geometric correction parameters for further processing.

The collected dataset serves as the foundation for preprocessing, thermal band extraction, radiometric corrections, and LST computation, enabling a comprehensive spatial analysis of temperature variations in the study area.

2.2 Band Used for Land Surface Temperature

For the **Land Surface Temperature (LST) analysis**, four spectral bands from Landsat Level 1 data were downloaded and utilized. These bands include:

- **Band 10**
- **Band 11**

- **Band 4**
- **Band 5**

Band 10 and 11 are **Thermal Infrared Sensor (TIRS)** for temperature calculations, as well as **Band 4 (Red) and Band 5 (Near-Infrared - NIR)** from the **Operational Land Imager (OLI)** for vegetation analysis through the **Normalized Difference Vegetation Index (NDVI)**.

In addition to the spectral bands, **metadata files** were obtained from the **United States Geological Survey (USGS) Earth Explorer**, which provided crucial radiometric and geometric correction parameters, thermal constants, and band-specific coefficients necessary for accurate calculations.

2.3 Time Series Analysis (2014 and 2024)

For this study, **Landsat Level 1 satellite imagery** was acquired for two different years **2014 and 2024** to analyze the temporal variations in **Land Surface Temperature (LST)** across **Offa Local Government Area (LGA)**. The datasets were carefully selected to ensure minimal cloud interference and optimal visibility for accurate thermal and vegetation analysi

1.) <u>2014 Landsat Data Acquisition</u>

The first dataset was acquired on **December 14, 2014**, during the **dry season** in Nigeria, which corresponds to the **harmattan period** in West Africa. During this time, the region experiences cooler temperatures at night but relatively warm daytime conditions due to the absence of rainfall and increased solar radiation. The **cloud cover for this dataset was exceptionally low at 0.05**, making it ideal for **Land Surface Temperature (LST) and Normalized Difference Vegetation Index (NDVI) analysis**.

Given that December falls within the **winter period** in the Northern Hemisphere, the influence of global seasonal variations might contribute to slightly lower atmospheric temperatures. However, in tropical regions like Nigeria, the dominant factor affecting LST during this period is **low humidity and increased surface heating due to clear skies**.

2.) <u>2024 Landsat Data Acquisition</u>

The second dataset was acquired on **November 23, 2024**, also during the **dry season** but at the beginning of the harmattan period. This season is marked by a gradual drop in humidity and rainfall, leading to **higher surface temperatures during the day** due to prolonged sunshine exposure. Unlike the 2014 dataset, this image had a **slightly higher cloud cover of 0.5**, which is still within acceptable limits for thermal analysis.

November in Nigeria is characterized by **transitioning weather patterns**, where the shift from the rainy season to the dry season results in increased solar radiation absorption by land surfaces. This can lead to higher **LST values compared to the wet season**, when increased vegetation and moisture would normally reduce surface heating.

Seasonal Impact on LST Analysis

The selection of data from similar seasonal periods (November and December) ensures consistency in **climatic conditions**, reducing the influence of seasonal temperature fluctuations on the LST comparison. The **dry season** is typically associated with **higher surface temperatures, reduced vegetation cover, and increased urban heat island (UHI) effects** due to lower soil moisture and minimal cloud coverage.

The data from **2014 and 2024** will be compared to assess **long-term changes in land surface temperature**, vegetation influence on urban heating, and the effects of urban expansion over time. The minimal cloud cover in both datasets ensures **high accuracy in thermal analysis**, making them reliable for understanding urban climate dynamics in **Offa LGA**.

2.4 Image Preprocessing steps

After acquiring the Landsat 8 data, the first preprocessing step involved I took is loading the dataset into **ArcGIS Pro** for analysis. Since **Land Surface Temperature (LST) estimation** relies on thermal infrared bands, **Band 10 and Band 11** from the Thermal Infrared Sensor **(TIRS)** were extracted and processed.

To minimize atmospheric noise and improve temperature accuracy, the **Cell Statistics** tool was used to calculate the **mean value** of the two bands. This method helps to **reduce sensor-related variations** and provides a more stable thermal representation of the Earth's surface. The output raster from this step served as the **base thermal layer** for subsequent radiometric corrections and LST computation.

This preprocessing step ensures that the final LST values are more **accurate and reliable**, contributing to a precise analysis of temperature variations in Offa LGA.

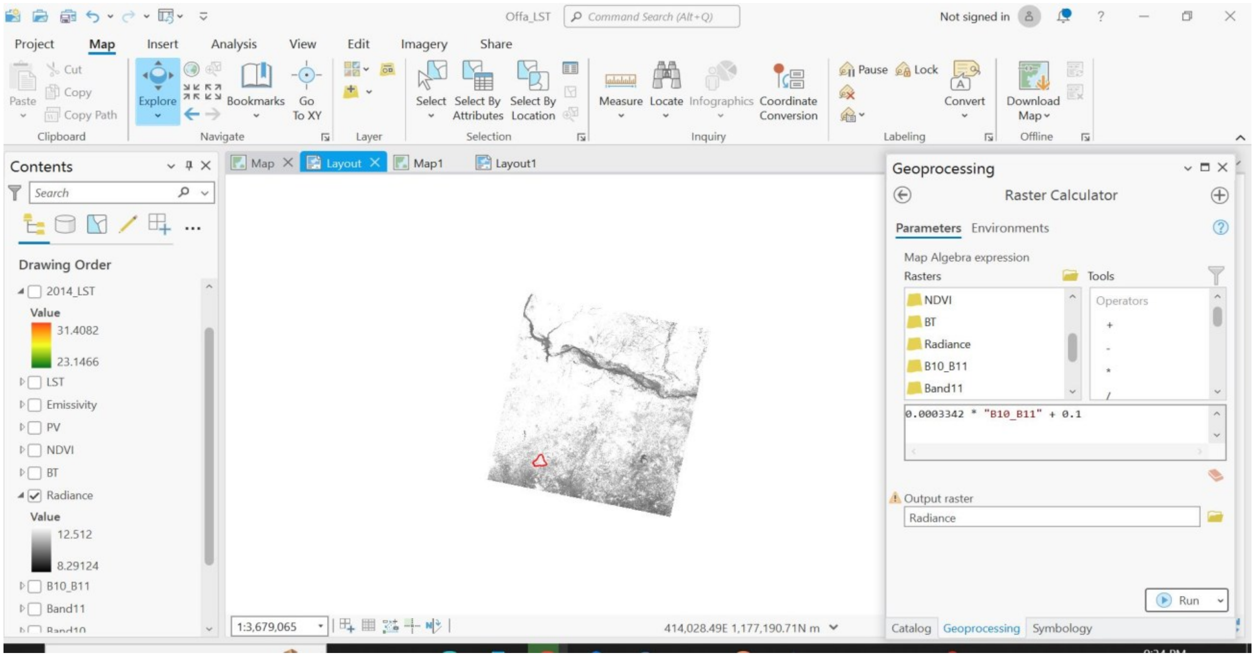
Subsequent Radiometric correction

Step 1: Convert Digital Numbers (DN) to Top of Atmosphere (TOA) Radiance

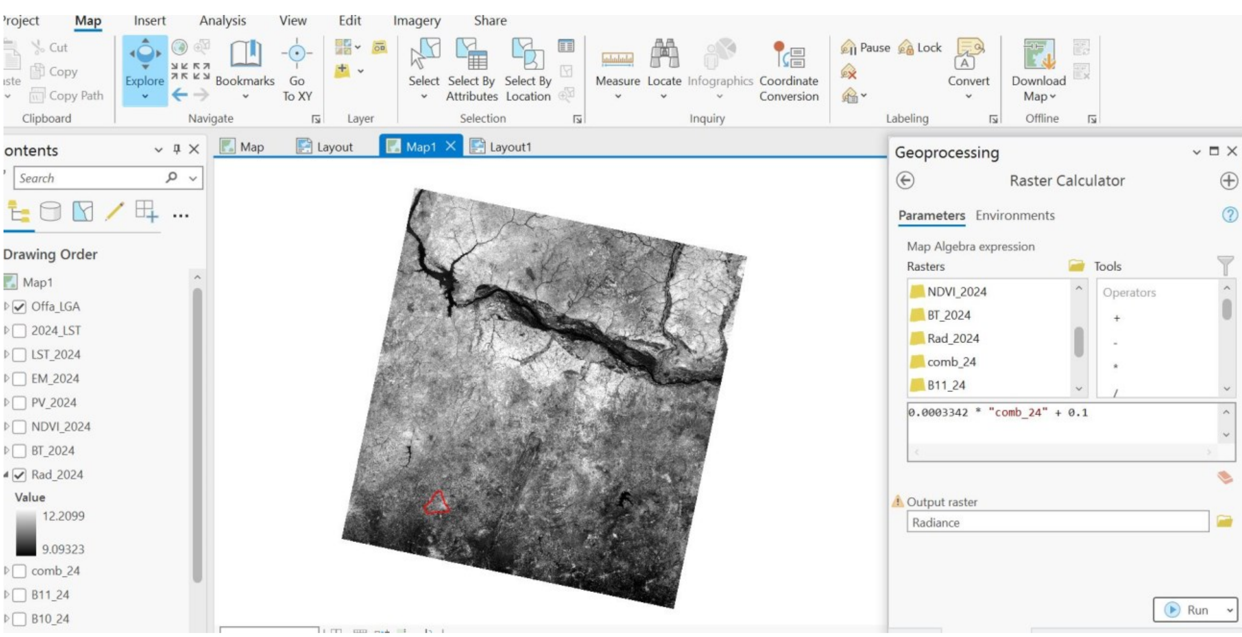
Landsat data is provided in Digital Number (DN) format, which needs to be converted to Top of Atmosphere (TOA) Radiance using the equation:

$$L\lambda = ML * Qcal + AL$$

- $L\lambda$ = TOA Spectral Radiance
- ML = Radiance Multiplier (from metadata)
- AL = Radiance Additive (from metadata)
- Qcal = Pixel Value (DN)



SPECTRAL RADIANCE OF 2014



SPECTRAL RADIANCE OF 2024

Step 2: Convert TOA Radiance to Brightness Temperature (BT)

After converting the **Digital Numbers (DN)** to **Top of Atmosphere (TOA) Radiance**, the next step I took in the **Land Surface Temperature (LST) analysis** is to compute the **Brightness Temperature (BT)**. This is done using the **thermal constants (K1 and K2)** provided in the metadata file. The formula used is:

$$\mathbf{BT = K2/In\ (k1/\ L\lambda + 1)}$$

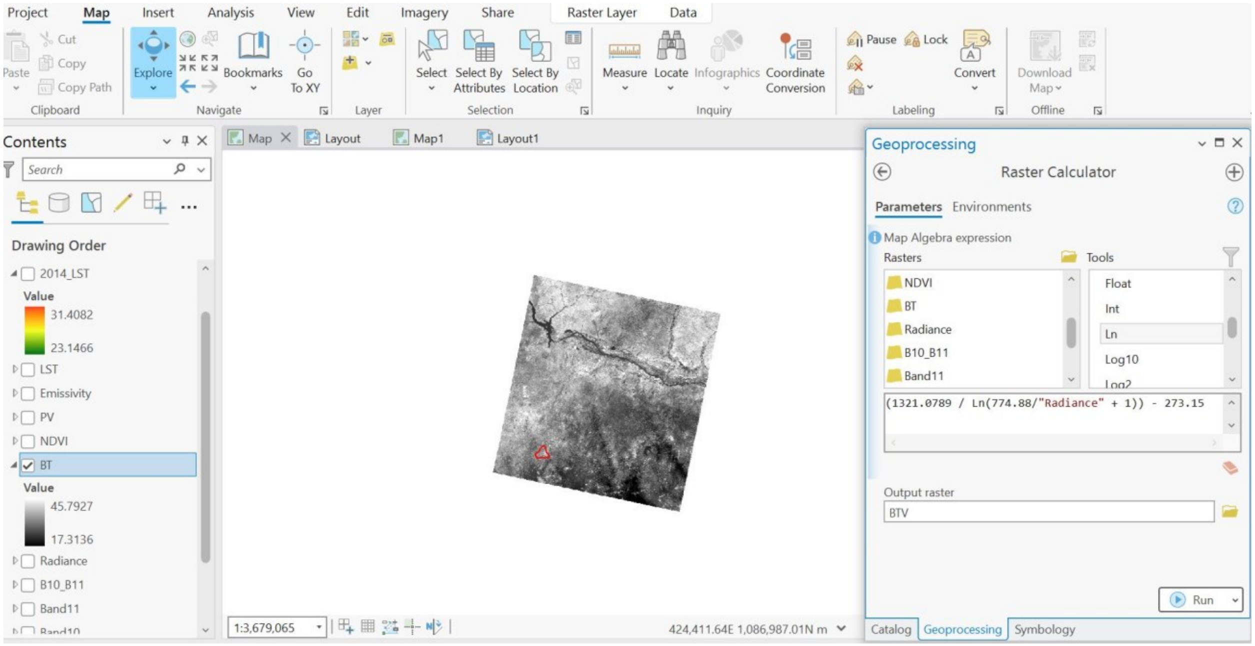
Where:

- BT = Brightness Temperature in **Kelvin (K)**
- K1 and K2 = Thermal conversion constants from the Landsat metadata
- $L\lambda$ = TOA Radiance

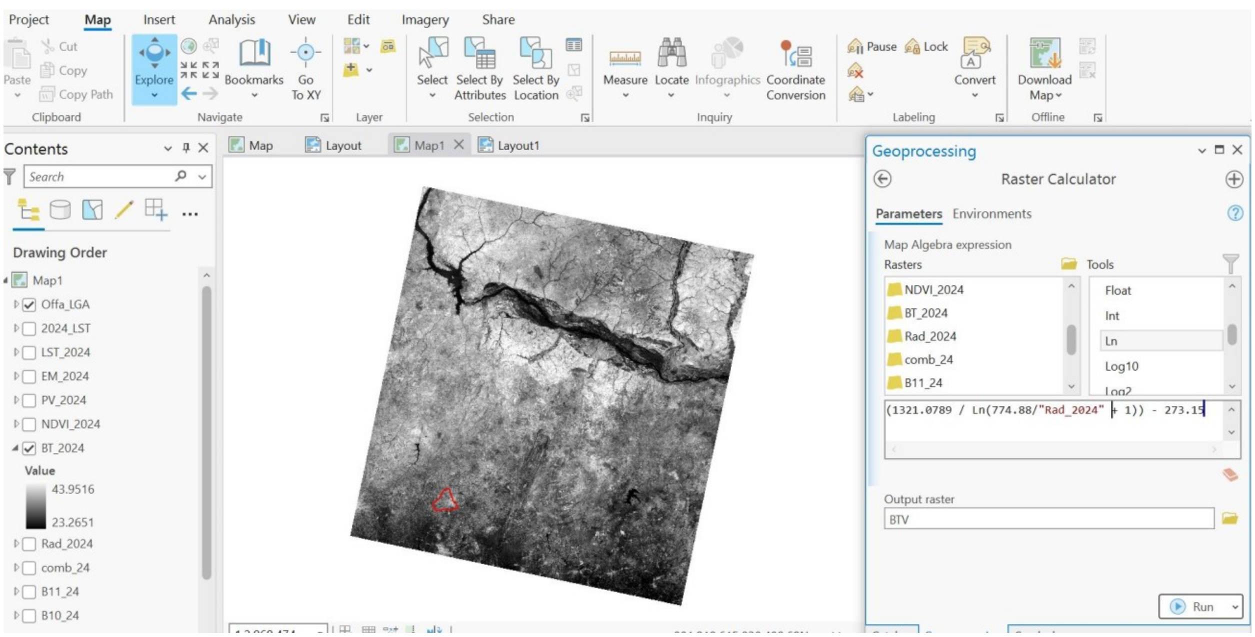
Since **Kelvin is not a common temperature unit for environmental analysis**, the final result was converted to **degrees Celsius (°C)** by subtracting **273.15** from the computed **BT values**:

$$\mathbf{BT(°C) = BT\ (K) - 273.15}$$

This conversion ensures that the temperature values are in a more interpretable unit for climate studies, environmental monitoring, and urban heat analysis in Offa LGA.



BRIGHTNESS TEMPERATURE VALUE IN CELCIUS 2014



BRIGHTNESS TEMPERATURE VALUE IN CELCIUS 2024

<u>Step 3</u>: Calculate Normalized Difference Vegetation Index (NDVI)

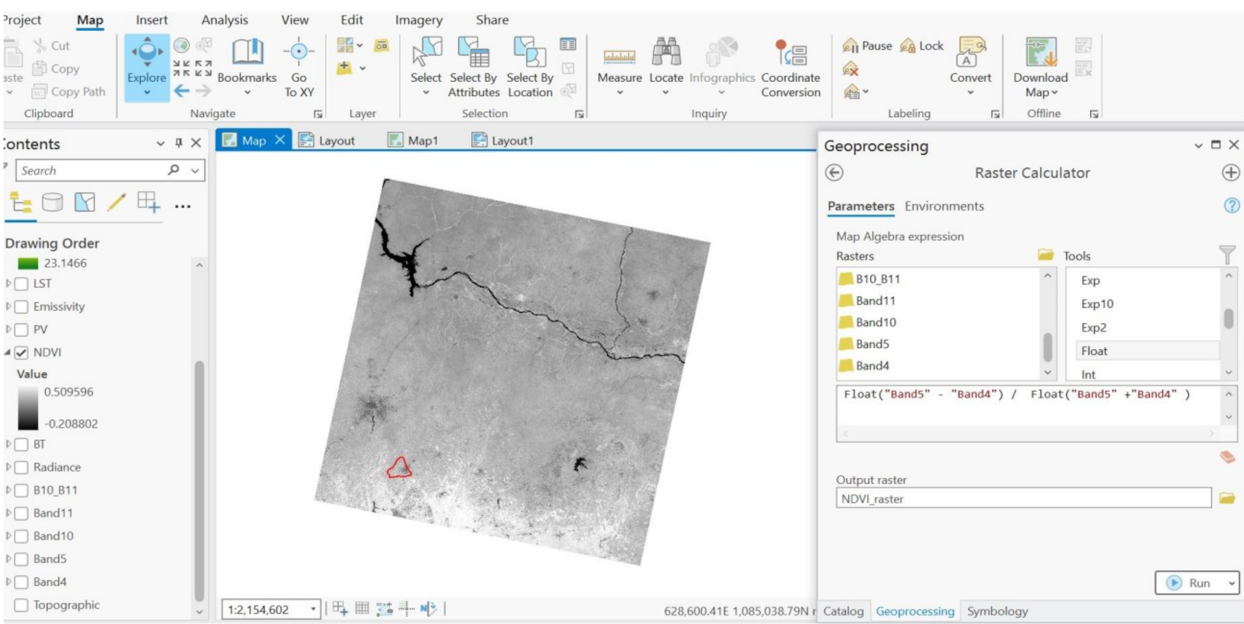
The **Normalized Difference Vegetation Index (NDVI)** was calculated to determine **vegetation cover** and estimate **land surface emissivity (LSE)**, which is crucial for accurate **Land Surface Temperature (LST) estimation**. NDVI was derived using **Band 4 (Red) and Band 5 (Near-Infrared - NIR)** from the Landsat dataset using the formula:

$$\mathbf{NDVI = (NIR + Red) / (NIR - Red)}$$

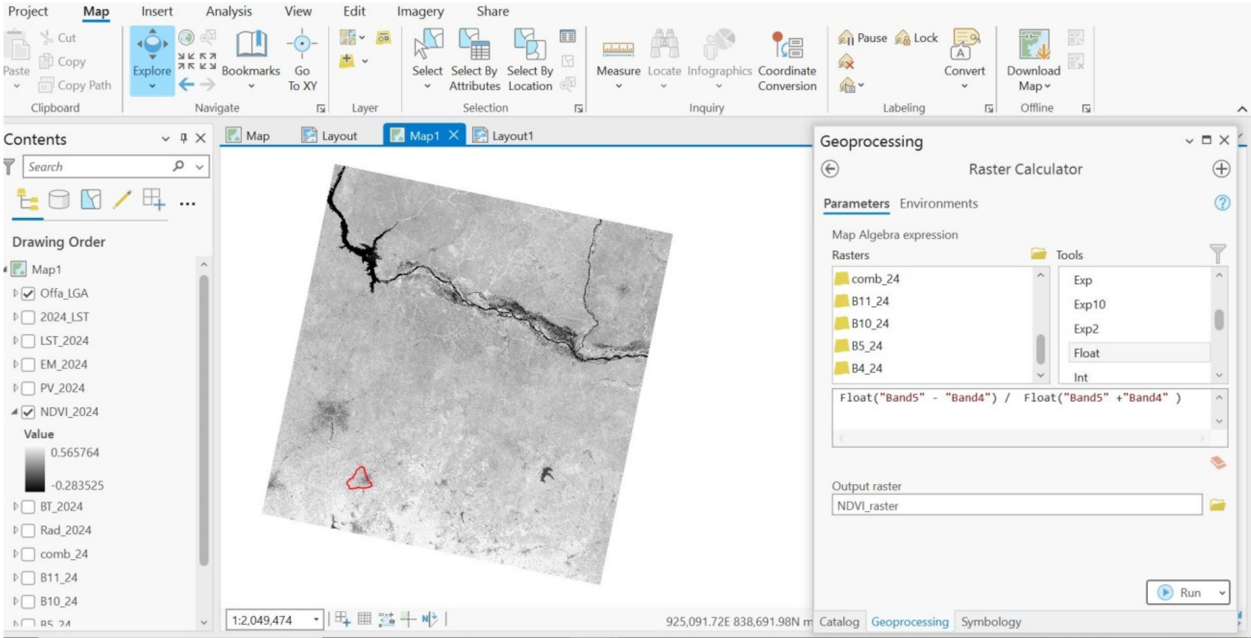
Where:

- **NIR (Band 5)** reflects strongly in healthy vegetation.
- **Red (Band 4)** is absorbed by plants but reflected by non-vegetated surfaces.

NDVI values range from **-1 to +1**, where higher values indicate **dense vegetation** and lower values indicate **bare surfaces or built-up areas**. The NDVI result was later used to compute **land surface emissivity (LSE).**



NDVI VALUE FOR YEAR 2014



NDVI FOR YEAR 2024

<u>Step 4</u>: Derived land surface Emssivity using Proportion of Vegetation

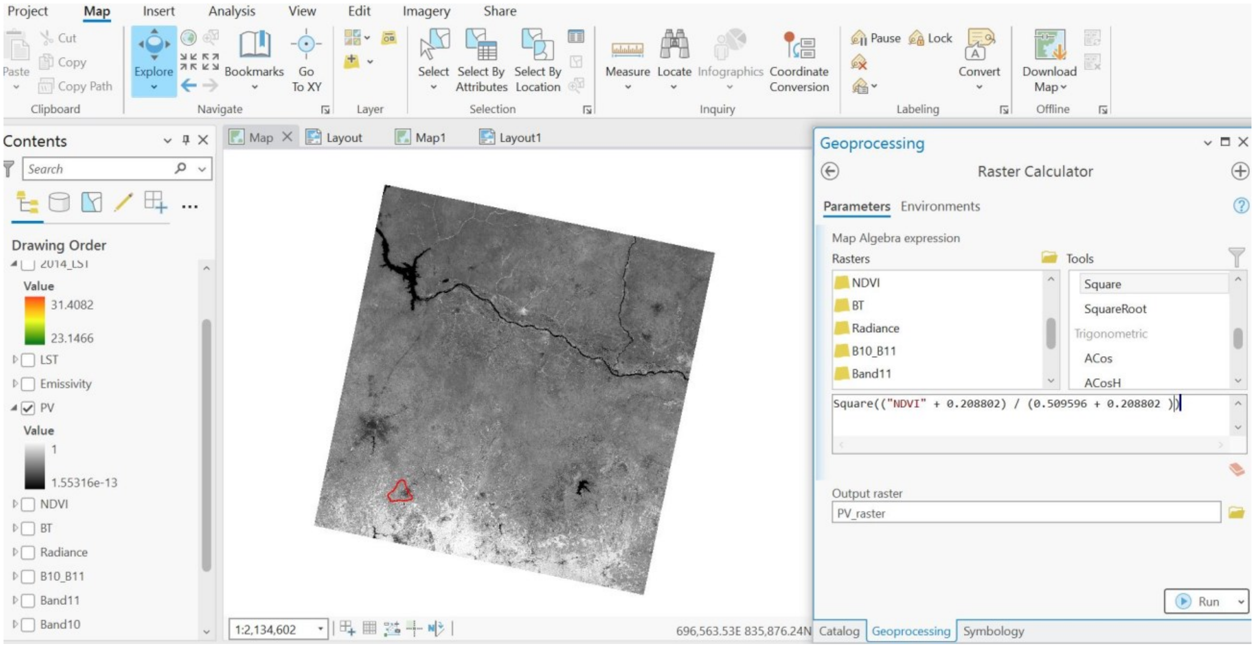
The **Proportion of Vegetation (Pv)** was derived from the **NDVI values** to estimate **Land Surface Emissivity (LSE)**, which is essential for accurate **Land Surface Temperature (LST) analysis**. The proportion of vegetation was calculated using the formula:

$$\mathbf{PV = ((NDVI - NDVI_{min}) / (NDVI_{max} - NDVI_{min}))^2}$$

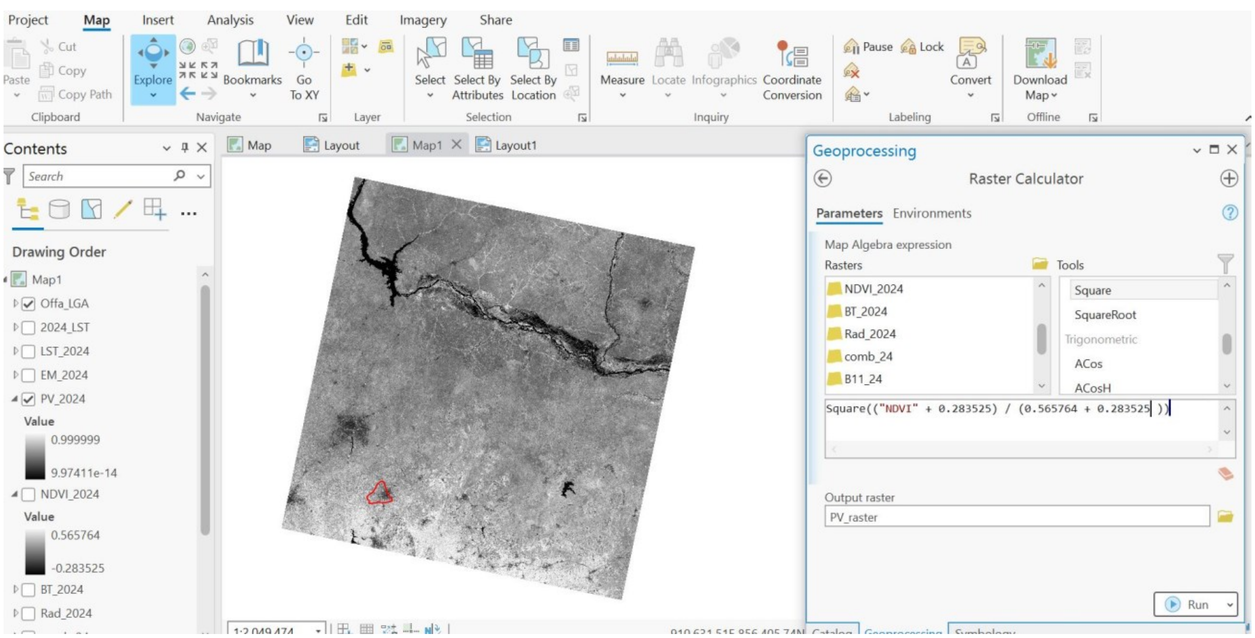
Where:

- NDVI = Normalized Difference Vegetation Index.
- NDVImax and NDVImin = Maximum and minimum NDVI values in the study area.

The **Pv** value ranges between **0 and 1**, where higher values indicate more vegetation cover and lower values indicate bare land or built-up areas. This step was crucial in estimating **Land Surface Emissivity (LSE)**, which directly affects the accuracy of LST calculations in Offa LGA.



PROPORTIONAL VEGETION VALUE YEAR 2014



PROPORTION VEGETION VALUE FOR YEAR 2024

<u>**Step 5:**</u> Land Surface Emissivity (LSE) Calculation

Land Surface Emissivity (LSE) was computed using the **Proportion of Vegetation (Pv) values**, as **LSE** is influenced by surface characteristics such as vegetation cover and land type. The formula used is:

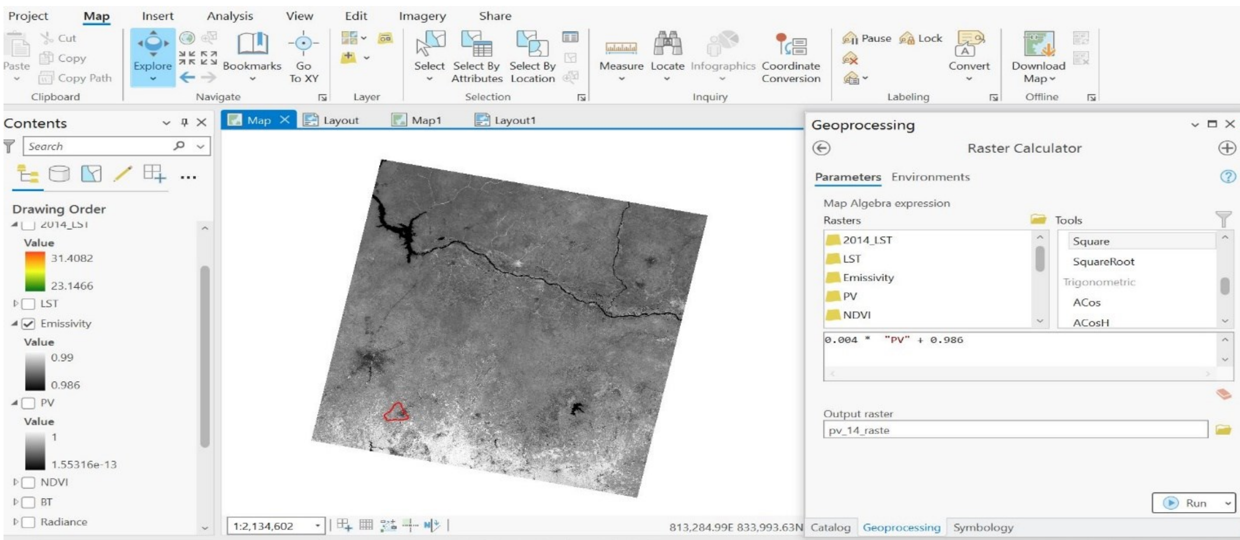
$$\boldsymbol{\varepsilon = 0.004PV + 0.986}$$

Ɛ = Land Surface Emissivity (LSE)

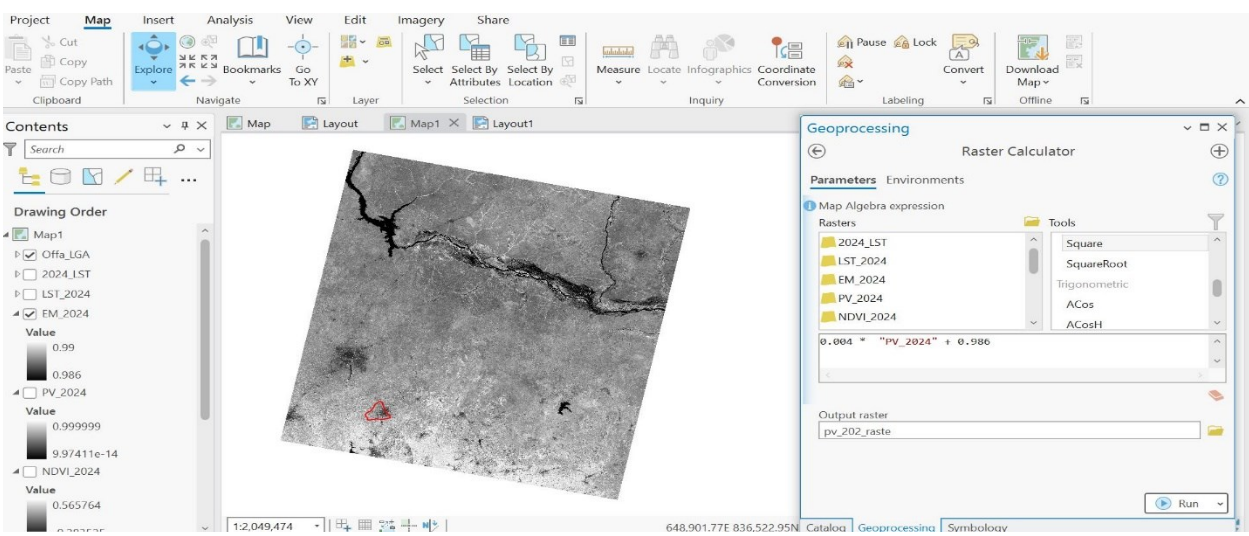
0.004 = The **0.004 coefficient** represents the **influence of vegetation density** on surface emissivity. It acts as a scaling factor that adjusts the emissivity based on the **NDVI values**.

0.986 = Correspond to a correction of the equation

PV = Proportion of Vegetation



EMISSIVITY VALUE OF YEAR 2014



EMISSIVITY VALUE OF YEAR 2024

CHAPTER THREE

LAND SURFACE TEMPERATURE OF OFFA LGA

3.1 Land Surface Calculation

After determining **Brightness Temperature (BT)** and **Land Surface Emissivity (LSE)**, the final step in the analysis was to compute **Land Surface Temperature (LST)** using the following equation:

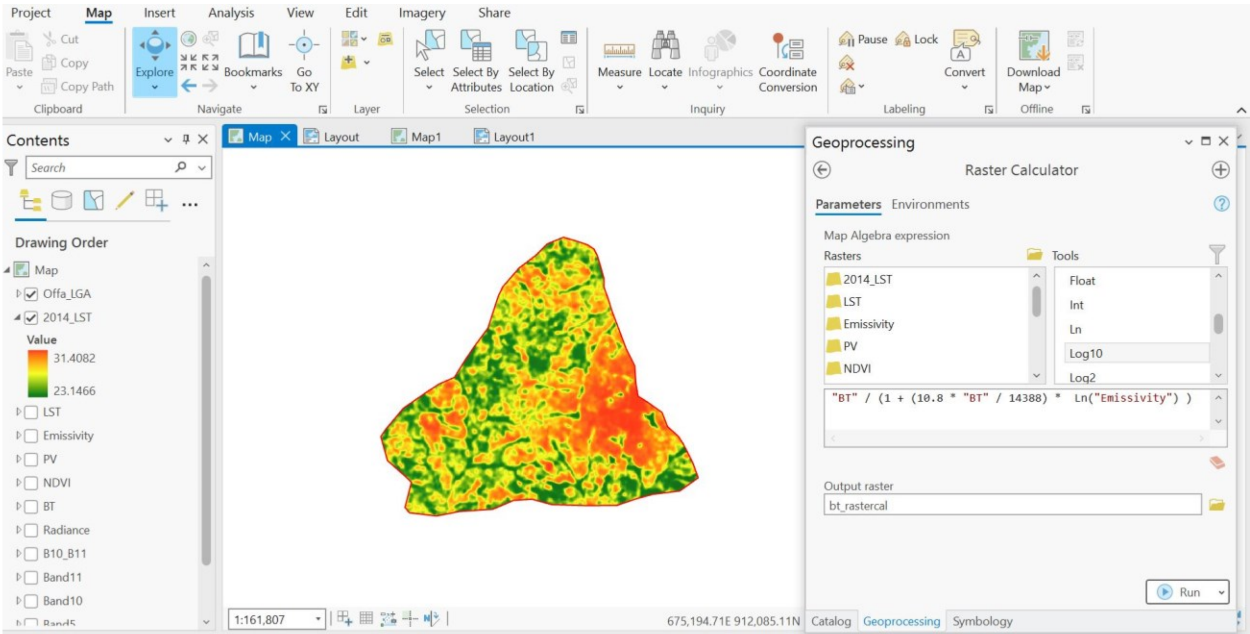
$$\mathbf{LST = BT/1 + (\lambda.BT/C_2)\ In(\varepsilon)}$$

Where:

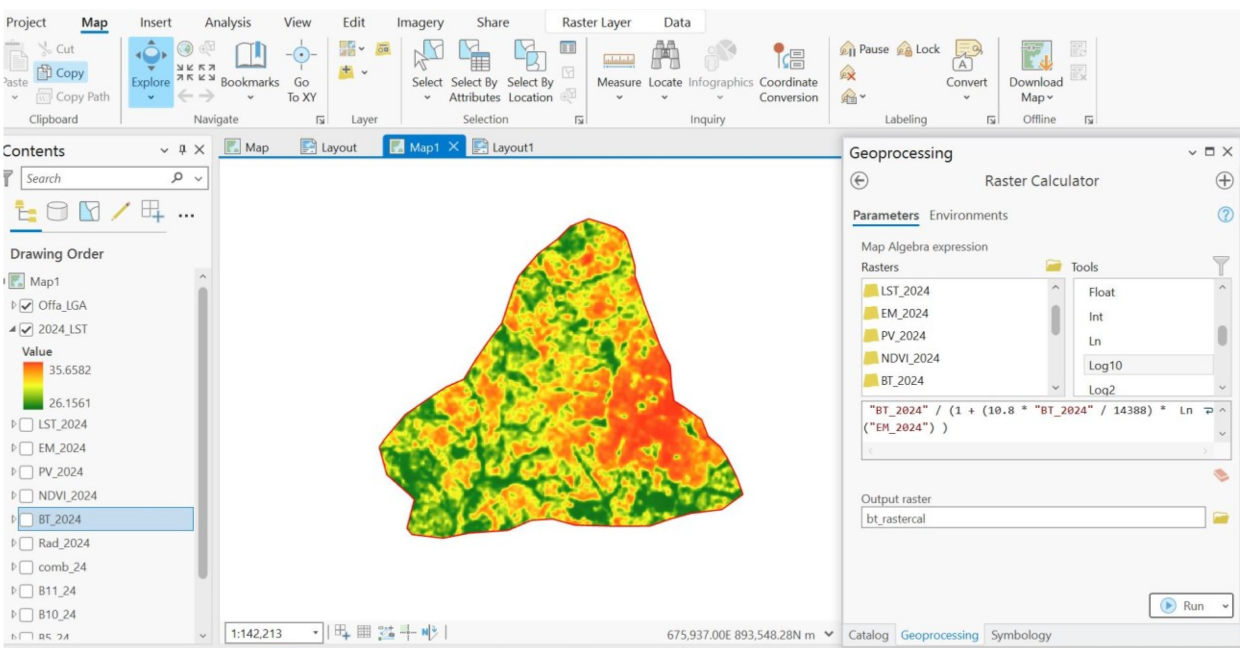
- **LST** = Land Surface Temperature in **Kelvin (K)**.
- **BT** = Brightness Temperature (computed earlier).
- **λ (wavelength)** = Effective wavelength of the thermal band (~10.895 µm for Landsat 8).
- **c_2** = Second radiation constant (1.438×10^{-2} m K).
- **ε** = Land Surface Emissivity (computed from NDVI)

Land Surface Temperature (LST) is a crucial parameter in environmental and climatic studies as it represents the thermal emission of the Earth's surface.

The final LST deliverables for the multi-date satellite imageries provides insights into **surface heat variations** across **Offa LGA**, allowing for analysis of urban heat islands, land cover impact, and temperature trends over time.



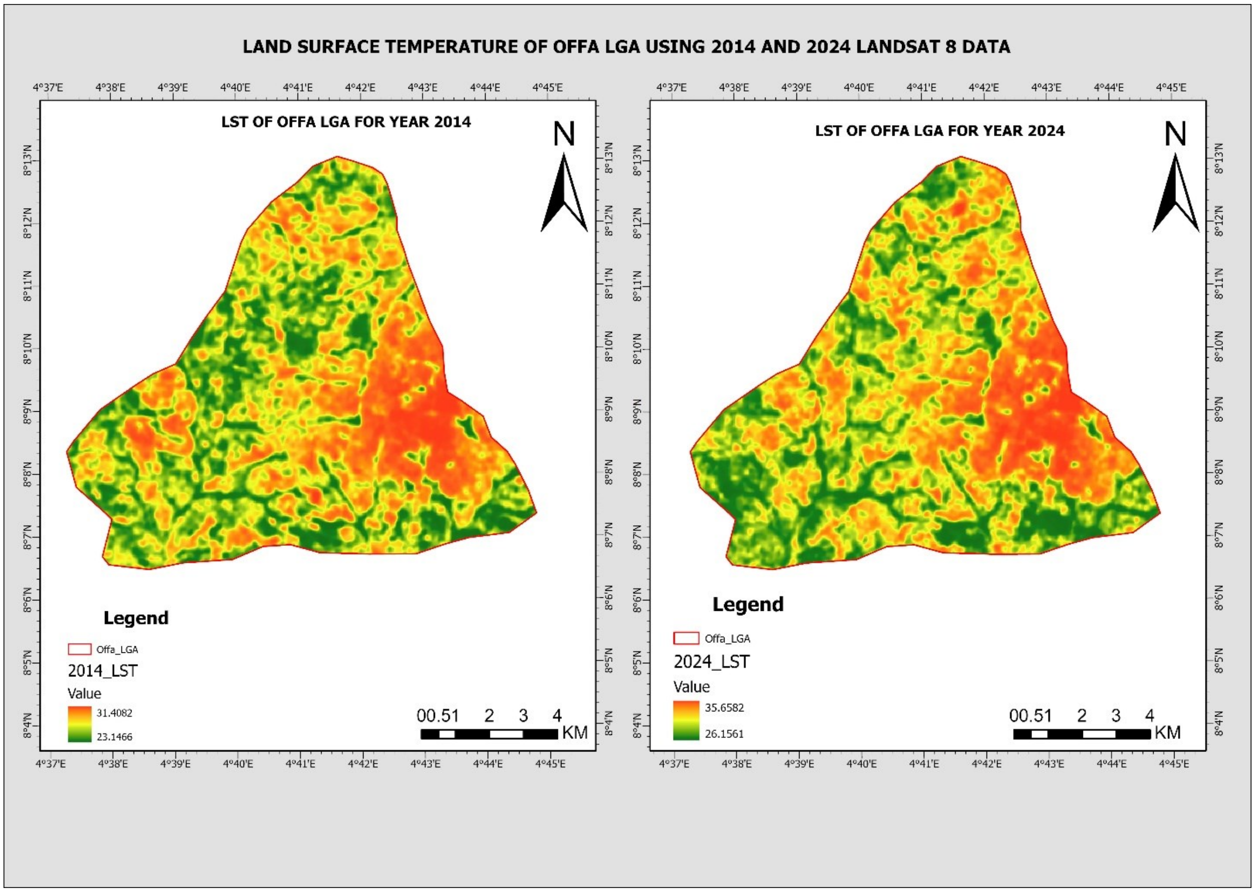
LST CALCULATION YEAR 2014



LST CALCULATION YEAR 2024

3.2 LST Classification and Visualization

To effectively analyze and compare the **Land Surface Temperature (LST)** variations in **Offa LGA**, the classified LST maps for the years **2014 and 2024** were generated using **Landsat 8 thermal bands**. The classification was based on temperature values extracted from the computed LST raster in ArcGIS Pro**.**



Classification Scheme

The LST values were categorized into distinct classes to highlight temperature variations across the study area. The classification was based on the minimum and maximum temperature ranges observed for each year:

- **2014 LST Range: 23.15°C – 31.41°C**
- **2024 LST Range: 26.16°C – 35.66°C**

The color gradient applied in the maps represents:

- **Green (Cooler Areas):** Areas with high vegetation cover or water bodies.
- **Yellow-Orange (Moderate Temperature):** Transitional zones between built-up and vegetated areas.
- **Red (Hotter Areas):** Urbanized or bare soil regions with high heat retention.

Visualization Approach

- **Side-by-Side Comparison:** The maps for 2014 and 2024 were displayed adjacent to facilitate visual assessment of temperature differences over time.
- **Consistent Color Scale:** A uniform color gradient was used for both years to ensure a direct comparison of LST distribution patterns.
- **Legend and Scale Bar:** To enhance interpretation, a **legend** was included to indicate temperature values, while a **scale bar** provided spatial reference.
- **Boundary Overlay:** The **Offa LGA boundary** was outlined to demarcate the study area clearly.

CHAPTER FOUR

RESULT AND ANALYSIS

4.1 LST Distribution in Offa LGA (2014 & 2014)

In **2014**, the LST values ranged from **23.15°C to 31.41°C**, with:

- **Cooler areas (green zones)** predominantly covering vegetated regions, indicating the presence of natural land cover that helps moderate surface temperature.
- **Moderate temperature areas (yellow-orange zones)** occurring in mixed land-use zones, including semi-urban settlements.
- **Hotter areas (red zones)** concentrated in the **southern and southeastern** parts of Offa LGA, corresponding to built-up and bare land surfaces, which absorb and retain heat more efficiently than vegetated areas.

LST Distribution in 2024

By **2024**, the temperature range increased to **26.16°C – 35.66°C**, showing a notable rise in overall LST values. Key observations include:

- A **shift towards higher temperatures**, indicating increasing land surface heating.
- An **expansion of red zones**, suggesting increased urbanization, deforestation, and impervious surfaces.
- Green zones remain but appear to be shrinking, likely due to vegetation loss.

4.2 Comparative analysis between 2014 – 2014 Data

- The **minimum LST increased by about 3°C**, while the **maximum LST increased by over 4°C** from 2014 to 2024.
- **Urban heat hotspots have expanded**, particularly in areas that were previously less affected.
- Vegetation cover, which helps in temperature regulation, has possibly declined, contributing to the **increase in LST**.

The results suggest that **urbanization, land-use change, and vegetation loss** are key factors influencing LST distribution in **Offa LGA** over the 10-year period. This highlights the need for sustainable urban planning and vegetation conservation to mitigate rising surface temperatures.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.1 CONCLUSION

The analysis of **Land Surface Temperature (LST) in Offa LGA** for the years **2014 and 2024** using **Landsat 8 data** reveals a significant increase in surface temperatures over the decade. The LST distribution shows that urbanized and bare land areas have experienced more intense warming, while vegetated regions have remained relatively cooler. The temperature increase, with a **rise of over 4°C in maximum LST**, indicates the growing impact of urbanization, land-use changes, and possibly climate variations. The observed expansion of **heat-prone zones** suggests an increase in impervious surfaces, such as roads and buildings, which absorb and retain more heat compared to vegetated areas.

This study highlights the importance of **remote sensing techniques** in monitoring land surface temperature variations over time. The results can be useful for **urban planning, environmental management, and climate adaptation strategies** to mitigate the adverse effects of rising temperatures in Offa LGA.

5. 2 RECOMMENDATION

1. **Urban Greening and Vegetation Conservation**
   Expanding green spaces through **afforestation, reforestation, and urban tree planting** can help reduce surface temperatures. Vegetation plays a crucial role in **heat absorption, evapotranspiration, and carbon sequestration**, making it a natural solution to mitigate rising LST.
2. **Sustainable Urban Planning and Land Use Management**
   Implementing **climate-sensitive infrastructure**, such as **cool roofing, reflective surfaces, and green rooftops**, can help minimize heat retention in built-up areas. Proper zoning regulations should be enforced to balance urban development with environmental sustainability, preventing excessive land conversion from vegetation to impervious surfaces.

3. **Remote Sensing and Climate Monitoring**
   Regular monitoring of **LST trends using satellite imagery and GIS technology** can help detect changes early and guide decision-making. Integrating **climate adaptation strategies** and public awareness programs will ensure communities are informed and actively participate in **sustainable environmental practices**.

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